Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$S=-\frac{1}{2}\int d\sigma d\tau[\partial_\mu\chi_1\partial^\mu\chi_1+2\chi_2\partial_\mu\chi_1\partial^\mu\chi_3+\frac{1}{1+{\chi_1}^2}\partial_\mu\chi_2\partial^\mu\chi_2+({\chi_2}^2+\frac{1}{1+{\chi_1}^2})\partial_\mu\chi_3\partial^\mu\chi_3+\partial_\mu t\partial^\mu t].$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\left({}^cD_{0+}^{\alpha,\alpha',\beta,\beta',\gamma}t^{\rho-1}\right)(x)=\frac{\Gamma(\rho)\Gamma(\alpha-\beta+\rho-m)\Gamma(\alpha+\alpha'+\beta'-\gamma+\rho-m)}{\Gamma(-\beta+\rho-m)\Gamma(\alpha+\alpha'-\gamma+\rho)\Gamma(\alpha+\beta'-\gamma+\rho-m)}x^{\alpha+\alpha'-\gamma+\rho-1}.$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\phi(\vec{X},\varepsilon)=\dot{\vec{X}}\cdot(\ddot{\vec{X}}\wedge\dddot{\vec{X}}\wedge\ldots\wedge\overset{(n)}{\vec{X}})=\det(\dot{\vec{X}},\ddot{\vec{X}},\dddot{\vec{X}},\ldots,\overset{(n)}{\vec{X}})=0$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$|p,\sigma,q,\lambda\colon f(p)\rangle=exp\Bigg[-e\int\sum_{\lambda=1,2}\frac{dk^+}{\sqrt{2k^+}}\int\frac{d^2k_\perp}{(2\pi)^{\frac{3}{2}}}\big[f(k,\lambda\colon p)a^\dagger(k,\lambda)-f^*(k,\lambda\colon p)a(k,\lambda)\big]\Bigg]|p,\sigma,q,\lambda\rangle\ .$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\Psi_2\left(\rho_1(y_1,y_1')^p,\boldsymbol{d}_{\mathcal{S}_2}(s_2,s_2')^p\right)=\Psi_2\circ\psi_2\left(\boldsymbol{d}_{\mathcal{Y}_1}(y_1,y_1')^p,\boldsymbol{d}_{\mathcal{Y}_2}(y_2,y_2')^p,\boldsymbol{d}_{\mathcal{X}}(x,x')^p\right).$$